Module 3

Measuring Corporate Performance

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

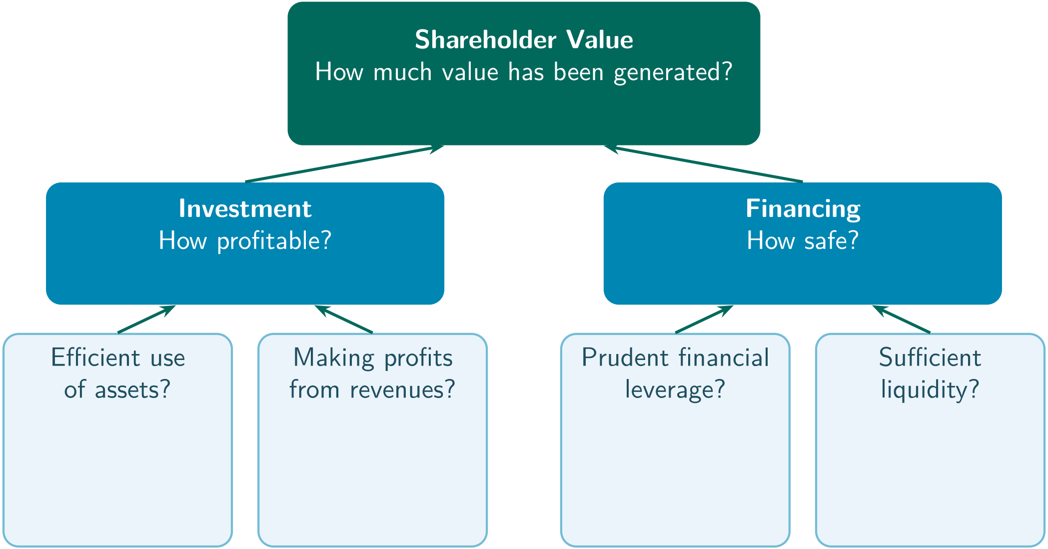
How Do We Evaluate a Corporation's Performance?

- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors

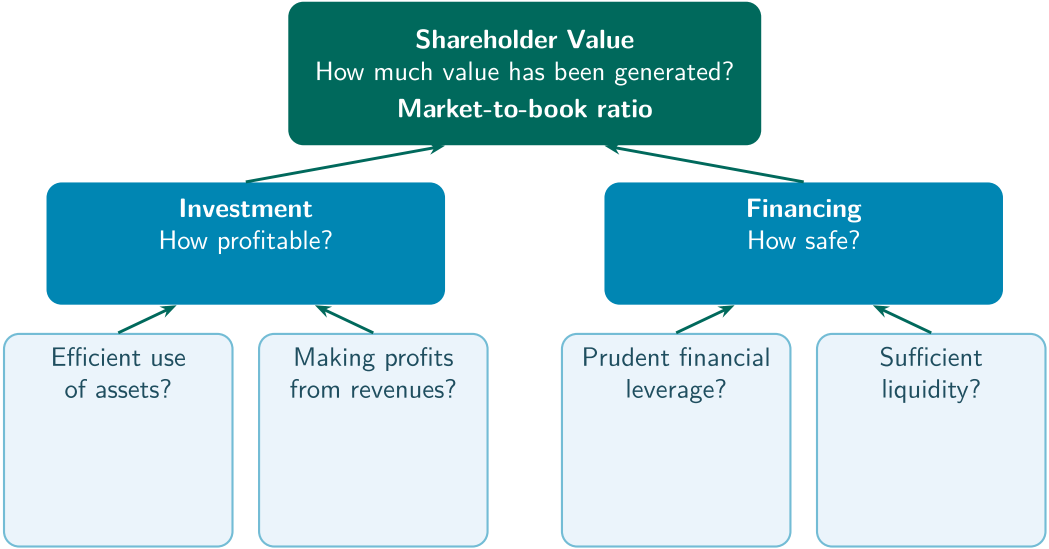
How Do We Evaluate a Corporation's Performance?

- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors
- Corporate performance is **multi-dimensional**

Measuring Corporate Performance



Measuring Corporate Performance



Shareholder Value

- Recall from **Module 2**, Market value of equity $=$ Market price of each share $\times$ Total shares outstanding

Shareholder Value

- Recall from **Module 2**, Market value of equity = Market price of each share × Total shares outstanding
- **Market capitalization**, another name for the market value of equity

Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn

"some kind of fruit company"

Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn
- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn

"some kind of fruit company"

Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn
- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn
- Market capitalization $=$ \$271 $\times$ 14,800 Mn $=$ \$4,010,800 Mn

"some kind of fruit company"

Shareholder Value, Market-to-Book Ratio

- **Market-to-book ratio**, the ratio of the market value of equity to the book value of equity

Shareholder Value, Market-to-Book Ratio

- **Market-to-book ratio**, the ratio of the market value of equity to the book value of equity

$$\text{Market-to-book} = \frac{\text{Market value of equity}}{\text{Book value of equity}}$$

Shareholder Value, Market-to-Book Ratio

- **Market-to-book ratio**, the ratio of the market value of equity to the book value of equity

$$\text{Market-to-book} = \frac{\text{Market value of equity}}{\text{Book value of equity}}$$

- From the previous slide, Apple's market-to-book ratio = \$4,010,800 Mn / \$73,733 Mn $\approx$ 54.4

Shareholder Value, Market-to-Book Ratio

- **Market-to-book ratio**, the ratio of the market value of equity to the book value of equity

$$\text{Market-to-book} = \frac{\text{Market value of equity}}{\text{Book value of equity}}$$

- From the previous slide, Apple's market-to-book ratio $=$ \$4,010,800 Mn / \$73,733 Mn $\approx$ 54.4

Pros

- Closely tied to how actual investors value the company

Cons

- Subject to market fluctuations

Practice Problem

- Here is a simplified balance sheet for a business (figures in $ millions). It has 600 million shares outstanding with a market price of $80 a share.

Current assets	$40,000	Current liabilities	$30,000
Fixed assets	50,000	Long-term liabilities	44,000
		Shareholders' equity	16,000
Total assets	**$90,000**	**Total liabilities and equity**	**$90,000**

- Calculate the market-to-book ratio.

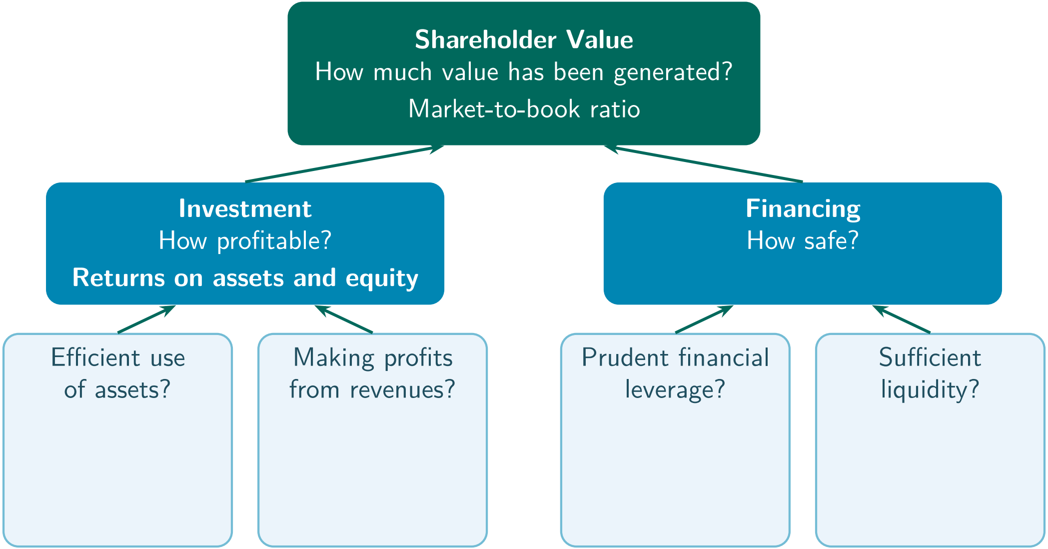
Practice Problem

- Here is a simplified balance sheet for a business (figures in \$ millions). It has 600 million shares outstanding with a market price of \$80 a share.

Current assets	\$40,000	Current liabilities	\$30,000
Fixed assets	50,000	Long-term liabilities	44,000
		Shareholders' equity	16,000
Total assets	**\$90,000**	**Total liabilities and equity**	**\$90,000**

- Calculate the market-to-book ratio. $600 \times \$80 \,/\, \$16{,}000 = \$48{,}000 \,/\, \$16{,}000 = 3.0$

Measuring Corporate Performance



Investment, Return on Resources

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets (book value)}}$$

Investment, Return on Resources

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets (book value)}}$$

$$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Shareholders' equity (book value)}}$$

Investment, Return on Resources

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets (book value)}}$$

$$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Shareholders' equity (book value)}}$$

Pros

- Less sensitive to stock market fluctuations

Cons

- Book values may deviate from market values

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
Taxes	840
Net income	**3,160**
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Example, Balance Sheet ($ Millions)

Assets		Liabilities and equity	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
Total fixed assets	29,200		
Total assets	**41,300**	**Total liabilities and equity**	**41,300**

Example, Return on Resources

$$\text{Return on assets} = \frac{\text{Net income}}{\text{Total assets}} = \frac{3{,}160}{41{,}300} = 0.077$$

Example, Return on Resources

$$\text{Return on assets} = \frac{\text{Net income}}{\text{Total assets}} = \frac{3{,}160}{41{,}300} = 0.077$$

$$\text{Return on equity} = \frac{\text{Net income}}{\text{Shareholders' equity}} = \frac{3{,}160}{19{,}900} = 0.159$$

Practice Problem

- Last year's ROE was 30%. This year the ROE has decreased to 20%, even though the firm's net income equaled last year's. What caused the decrease?

a. Shareholders' equity decreased by 10%.
b. Shareholders' equity decreased by 50%.
c. Shareholders' equity increased by 10%.
d. Shareholders' equity increased by 50%.

Practice Problem

- Last year's ROE was 30%. This year the ROE has decreased to 20%, even though the firm's net income equaled last year's. What caused the decrease?

a. Shareholders' equity decreased by 10%.
b. Shareholders' equity decreased by 50%.
c. Shareholders' equity increased by 10%.
d. Shareholders' equity increased by 50%.

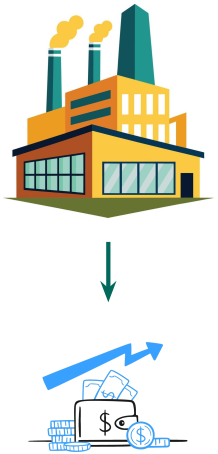
- Suppose net income is X in both years. Then 30% $= X$ / shareholders' equity last year, and 20% $= X$ / shareholders' equity this year
- Then $X =$ 30% $\times$ shareholders' equity last year $=$ 20% $\times$ shareholders' equity this year
- So shareholders' equity this year / shareholders' equity last year $=$ 30%/20% $=$ 150%

Measuring Corporate Performance

- **Shareholder Value**
  How much value has been generated?
  Market-to-book ratio
  - **Investment**
    How profitable?
    Returns on assets and equity
    - Efficient use of assets?
      Asset and inventory turnover
    - Making profits from revenues?
  - **Financing**
    How safe?
    - Prudent financial leverage?
    - Sufficient liquidity?

Efficiency, Use of Assets

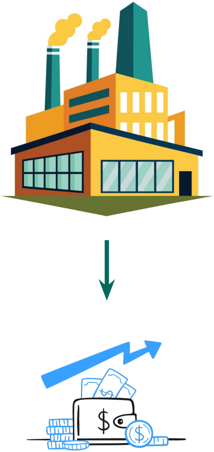
$$\text{Asset turnover ratio} = \frac{\text{Revenues}}{\text{Total assets}}$$



Efficiency, Use of Assets

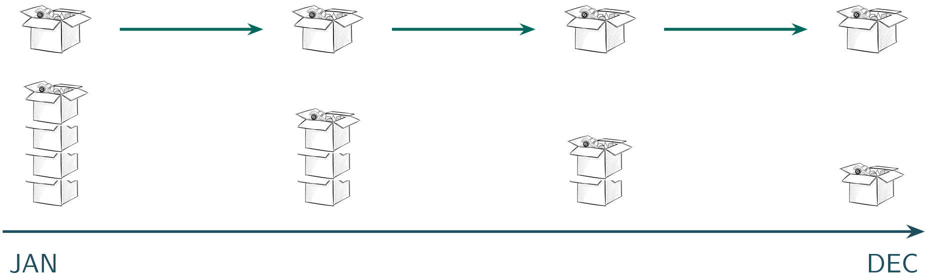
$$\text{Asset turnover ratio} = \frac{\text{Revenues}}{\text{Total assets}}$$

- How much in revenues is generated by each dollar of assets?



Efficiency, Inventory

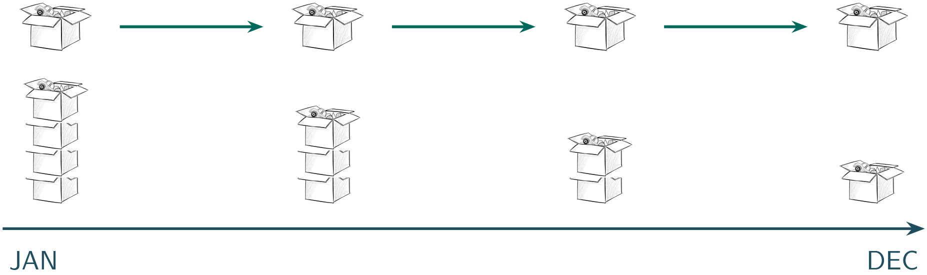
$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$



Efficiency, Inventory

$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

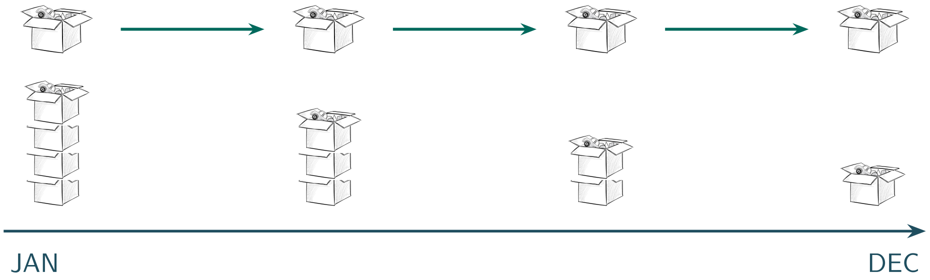
- Does the company turn over inventories rapidly?
- Efficient firms do not let inventories sit idle



Efficiency, Inventory

$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

- Does the company turn over inventories rapidly?
- Efficient firms do not let inventories sit idle
- Aim for the highest possible? No, inadequate inventories may fail to meet customer demand in time



Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales

Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales
  - Accounts receivable is from previous 7 days' sales, \$7,000

Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales
  - Accounts receivable is from previous 7 days' sales, \$7,000

$$\text{Average collection period} = \frac{\text{Accounts receivable}}{\text{Average daily revenues}}$$

Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales
  - Accounts receivable is from previous 7 days' sales, \$7,000

$$\text{Average collection period} = \frac{\text{Accounts receivable}}{\text{Average daily revenues}}$$

- The average length of time for customers to pay their bills

Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales
  - Accounts receivable is from previous 7 days' sales, \$7,000

$$\text{Average collection period} = \frac{\text{Accounts receivable}}{\text{Average daily revenues}}$$

- The average length of time for customers to pay their bills
- Efficient firms collect bills quickly

Efficiency, Accounts Receivable

- Suppose a firm has daily sales of \$1,000, and customers always repay 7 days after sales
  - Accounts receivable is from previous 7 days' sales, \$7,000

$$\text{Average collection period} = \frac{\text{Accounts receivable}}{\text{Average daily revenues}}$$

- The average length of time for customers to pay their bills
- Efficient firms collect bills quickly
- Aim for the fastest possible? No, it may reflect an over-stringent credit policy
  - **Credit policy**, the terms on which a firm lets customers buy now and pay later

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
Taxes	840
Net income	**3,160**
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Example, Balance Sheet ($ Millions)

Assets		Liabilities and equity	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
Total fixed assets	29,200		
Total assets	**41,300**	**Total liabilities and equity**	**41,300**

Example, Efficiency Ratios

$$\text{Asset turnover ratio} = \frac{80{,}000}{41{,}300} = 1.94$$

Example, Efficiency Ratios

$$\text{Asset turnover ratio} = \frac{80{,}000}{41{,}300} = 1.94$$

$$\text{Inventory turnover ratio} = \frac{56{,}000}{9{,}000} = 6.2$$

Example, Efficiency Ratios

$$\text{Asset turnover ratio} = \frac{80{,}000}{41{,}300} = 1.94$$

$$\text{Inventory turnover ratio} = \frac{56{,}000}{9{,}000} = 6.2$$

$$\text{Average collection period} = \frac{500}{80{,}000/365} = 2.3 \text{ days}$$

Practice Problem

- A firm had accounts receivable of $940 at the beginning of the year and generated $3,400 of revenues during the year. How much is its average collection period?

a. 94 days

b. 95 days

c. 101 days

d. 107 days

Practice Problem

- A firm had accounts receivable of $940 at the beginning of the year and generated $3,400 of revenues during the year. How much is its average collection period?

a. 94 days

b. 95 days

c. 101 days

d. 107 days

- Average collection period = 940 / (3,400 / 365) = 101 days

Measuring Corporate Performance

Shareholder Value
How much value has been generated?
Market-to-book ratio

Investment
How profitable?
Returns on assets and equity

Efficient use of assets?
Asset and inventory turnover

Making profits from revenues?
Profit margin

Financing
How safe?

Prudent financial leverage?

Sufficient liquidity?

Profits from Revenues, Profit Margin

$$\text{Profit margin} = \frac{\text{Net income}}{\text{Revenues}}$$

Profits from Revenues, Profit Margin

$$\text{Profit margin} = \frac{\text{Net income}}{\text{Revenues}}$$

- How much of each dollar of revenues ends up as net income?

Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?

Aswath Damodaran, "Dean of Valuation"

Excess Returns, January 22, 2026 (1:23), "Could AI Actually Reduce Profit Margins? Aswath Damodaran Thinks So"

Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?
- Do you agree?

Aswath Damodaran, "Dean of Valuation"

Excess Returns, January 22, 2026 (1:23), "Could AI Actually Reduce Profit Margins? Aswath Damodaran Thinks So"

Decompose ROA

- We can separate ROA into component ratios

Decompose ROA

- We can separate ROA into component ratios

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}}$$

Decompose ROA

- We can separate ROA into component ratios

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}}$$

$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}}$$

Decompose ROA

- We can separate ROA into component ratios

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}}$$

$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}}$$

$$\text{ROA} = \text{Asset turnover} \times \text{Profit margin}$$

Decompose ROA

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}} = \frac{3{,}160}{41{,}300} = 0.077$$

Decompose ROA

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}} = \frac{3{,}160}{41{,}300} = 0.077$$

$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}} = \text{Asset turnover} \times \text{Profit margin} = 1.94 \times 0.0395 = 0.077$$

Decompose ROA

Company	Strategy	Asset Turnover	Profit Margin	ROA
Zara	Fast fashion	1.6	7.5%	12%
Hermès	Luxury fashion	0.55	22.5%	12%

Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?

a. A current ratio of 1.6.
b. A P/E ratio of 14.
c. A return on equity of 25%.
d. A profit margin of 6.4%.

Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?

a. A current ratio of 1.6.

b. A P/E ratio of 14.

c. A return on equity of 25%.

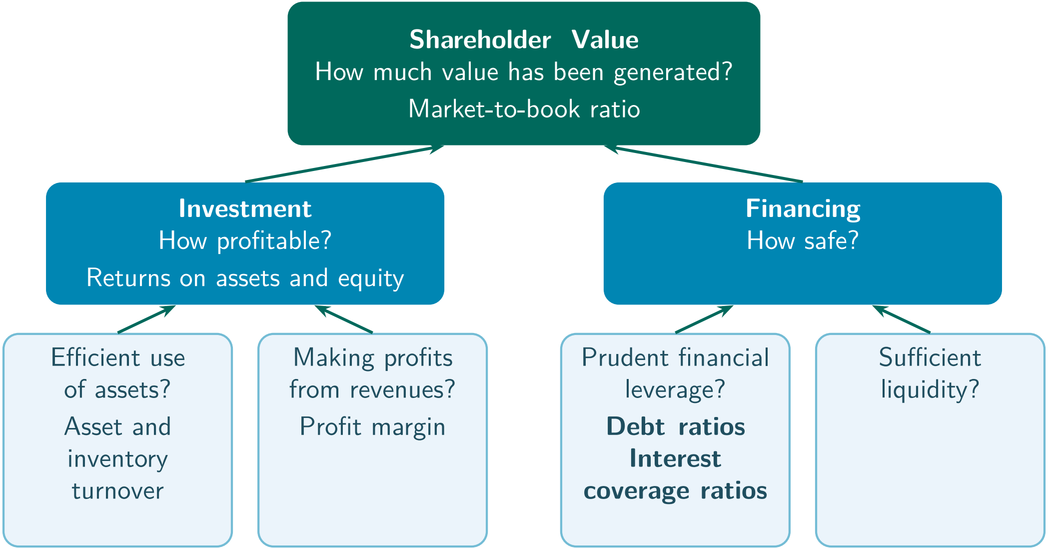
d. A profit margin of 6.4%.

- ROA $=$ Asset turnover $\times$ Profit margin, and $2.5 \times 6.4\% = 16\%$

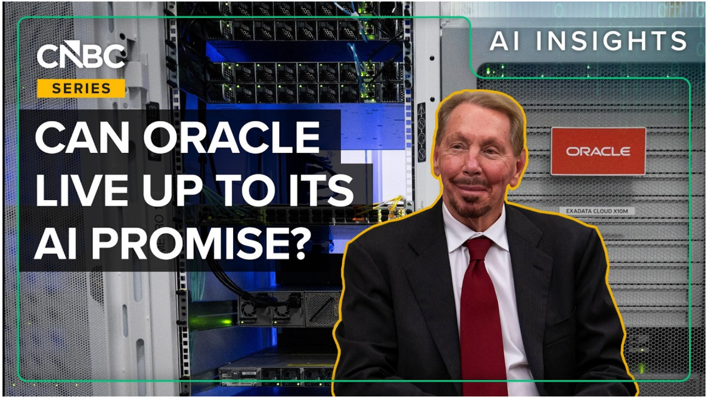
Recap, Investment

- Investment profitability is measured with the returns on assets and equity
- Turnover ratios, how much in revenues the assets generate and how quickly inventories sell
- Profit margin, how much of each dollar of revenues ends up as net income
- ROA $=$ Asset turnover $\times$ Profit margin

Measuring Corporate Performance



Motivating Example, Oracle



CNBC, March 10, 2026 (5:45), "How Oracle's AI-Fueled Debt Load Has Investors On Edge"

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good	**Bad**
\$6B (200%)	\$2B (0%)

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good
\$6B (200%)

Bad
\$2B (0%)

\$1B equity + \$1B debt

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good
\$6B (200%)

Bad
\$2B (0%)

\$1B equity + \$1B debt

Good

Bad

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

- **Good**: \$6B (200%)
- **Bad**: \$2B (0%)

\$1B equity + \$1B debt

- **Good**: Equity holders \$4.5B (350%)
- **Bad**

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good

\$6B (200%)

Bad

\$2B (0%)

\$1B equity + \$1B debt

Good

Equity holders	\$4.5B (350%)
Debt holders	\$1.5B (50%)

Bad

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good
\$6B (200%)

Bad
\$2B (0%)

\$1B equity + \$1B debt

Good

Equity holders	\$4.5B (350%)
Debt holders	\$1.5B (50%)

Bad

Equity holders	\$0.5B (−50%)

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good	Bad
\$6B (200%)	\$2B (0%)

\$1B equity + \$1B debt

Good		Bad	
Equity holders	\$4.5B (350%)	Equity holders	\$0.5B (−50%)
Debt holders	\$1.5B (50%)	Debt holders	\$1.5B (50%)

Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (good, 50% chance) or \$2B (bad, 50% chance)
- Payments to debt holders have *priority* over payments to equity holders
- Assume the interest rate on debt is 50%

All equity financed

Good	**Bad**
\$6B (200%)	\$2B (0%)

\$1B equity + \$1B debt

Good		**Bad**	
Equity holders	\$4.5B (350%)	Equity holders	\$0.5B (−50%)
Debt holders	\$1.5B (50%)	Debt holders	\$1.5B (50%)

With debt, gains and losses to equity holders are amplified

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand
- Expansion succeeds and profits rise $\rightarrow$ debt holders still receive only the fixed interest and principal, the gains go to shareholders

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand
- Expansion succeeds and profits rise $\rightarrow$ debt holders still receive only the fixed interest and principal, the gains go to shareholders
- Expansion fails and profits drop $\rightarrow$ debt holders are paid first, shareholders may be left with nothing

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand
- Expansion succeeds and profits rise $\rightarrow$ debt holders still receive only the fixed interest and principal, the gains go to shareholders
- Expansion fails and profits drop $\rightarrow$ debt holders are paid first, shareholders may be left with nothing
- Debt increases returns to shareholders in good times and reduces them in bad times

Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand
- Expansion succeeds and profits rise $\rightarrow$ debt holders still receive only the fixed interest and principal, the gains go to shareholders
- Expansion fails and profits drop $\rightarrow$ debt holders are paid first, shareholders may be left with nothing
- Debt increases returns to shareholders in good times and reduces them in bad times
  - This is **financial leverage**

Financial Leverage

$$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Equity}}$$

Financial Leverage

$$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Equity}}$$

$$\text{Long-term debt-to-equity ratio} = \frac{\text{Long-term debt}}{\text{Equity}}$$

Financial Leverage

$$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Equity}}$$

$$\text{Long-term debt-to-equity ratio} = \frac{\text{Long-term debt}}{\text{Equity}}$$

$$\text{Total debt ratio} = \frac{\text{Total liabilities}}{\text{Total assets}}$$

Example, Balance Sheet ($ Millions)

Assets		Liabilities and equity	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
Total fixed assets	29,200		
Total assets	**41,300**	**Total liabilities and equity**	**41,300**

Example, Leverage

$$\text{Long-term debt ratio} = \frac{11{,}300}{11{,}300 + 19{,}900} = 0.36$$

Example, Leverage

$$\text{Long-term debt ratio} = \frac{11{,}300}{11{,}300 + 19{,}900} = 0.36$$

$$\text{Long-term debt-to-equity ratio} = \frac{11{,}300}{19{,}900} = 0.57$$

Example, Leverage

$$\text{Long-term debt ratio} = \frac{11{,}300}{11{,}300 + 19{,}900} = 0.36$$

$$\text{Long-term debt-to-equity ratio} = \frac{11{,}300}{19{,}900} = 0.57$$

$$\text{Total debt ratio} = \frac{\text{Total liabilities}}{\text{Total assets}} = \frac{21{,}400}{41{,}300} = 0.52$$

Financial Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest expense}}$$

Financial Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest expense}}$$

$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest expense}}$$

Example, Income Statement ($ Millions)

	$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
Taxes	840
Net income	**3,160**
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Example, Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest payments}} = \frac{5{,}000}{1{,}000} = 5.0$$

Example, Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest payments}} = \frac{5{,}000}{1{,}000} = 5.0$$

$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest payments}} = \frac{5{,}000 + 3{,}000}{1{,}000} = 8.0$$

Practice Problem

- A firm has a times interest earned ratio of 2, a tax liability of \$1 million, and interest expense of \$1.5 million. Its book value of equity is \$1.5 million. How much is its ROE?

a. 26.67%

b. 30.00%

c. 33.33%

d. 50.00%

Practice Problem

- A firm has a times interest earned ratio of 2, a tax liability of \$1 million, and interest expense of \$1.5 million. Its book value of equity is \$1.5 million. How much is its ROE?

a. 26.67%

b. 30.00%

c. 33.33%

d. 50.00%

- Times interest earned = EBIT / Interest expense = 2, and interest expense = 1.5, so EBIT = 3
- Net income = EBIT − interest expense − taxes = 3 − 1.5 − 1 = 0.5
- ROE = Net income / Equity = 0.5 / 1.5 = 33.33%

Practice Problem

- A firm pays an 8% rate of interest on $10 million of outstanding debt with face value $10 million. The firm's EBIT was $1 million.

a. What is its times interest earned?

b. If depreciation is $200,000, what is its cash coverage ratio?

Practice Problem

- A firm pays an 8% rate of interest on \$10 million of outstanding debt with face value \$10 million. The firm's EBIT was \$1 million.

a. What is its times interest earned?
   Interest expense $= 0.08 \times$ \$10 million $=$ \$800,000
   Times interest earned $=$ \$1,000,000 / \$800,000 $= 1.25$
b. If depreciation is \$200,000, what is its cash coverage ratio?

Practice Problem

- A firm pays an 8% rate of interest on \$10 million of outstanding debt with face value \$10 million. The firm's EBIT was \$1 million.

a. What is its times interest earned?
   Interest expense $=$ 0.08 $\times$ \$10 million $=$ \$800,000
   Times interest earned $=$ \$1,000,000 / \$800,000 $=$ 1.25
b. If depreciation is \$200,000, what is its cash coverage ratio?
   Cash coverage ratio $=$ (\$1,000,000 $+$ \$200,000) / \$800,000 $=$ 1.5

Practice Problem

- Which one of the following will increase a firm's times interest earned ratio?

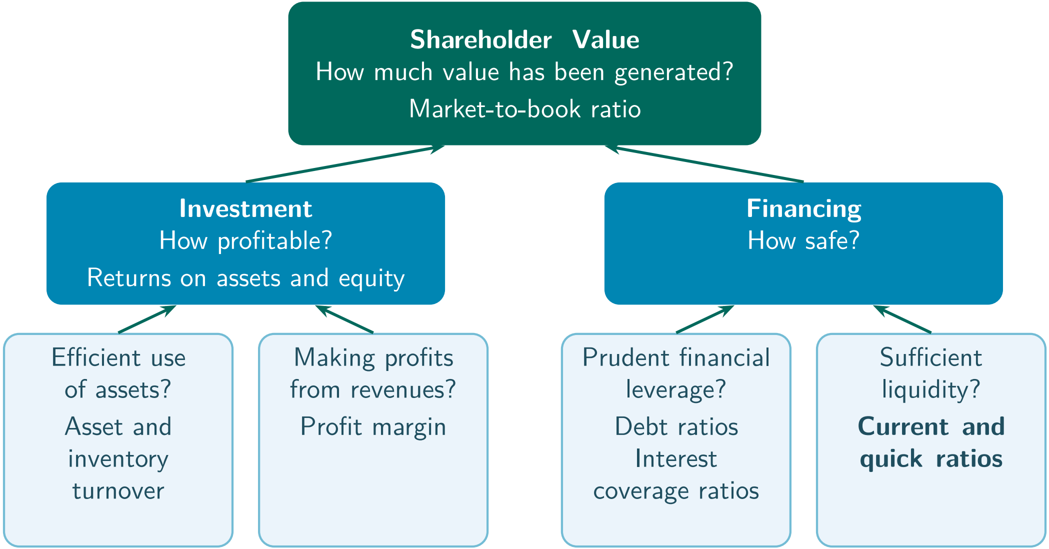
a. An increase in debt.
b. A decrease in cost of goods sold.
c. An increase in interest expense.
d. A decrease in net income.

Practice Problem

- Which one of the following will increase a firm's times interest earned ratio?

a. An increase in debt.
b. A decrease in cost of goods sold.
c. An increase in interest expense.
d. A decrease in net income.

Measuring Corporate Performance



Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value

Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value
- Liquid assets can be converted into cash quickly and cheaply

Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value
- Liquid assets can be converted into cash quickly and cheaply
- Liquidity metrics compare the amount of liquid assets to the amount of debt that requires immediate repayment

Example, Balance Sheet ($ Millions)

Most liquid ↓ Least liquid

Assets		Liabilities and equity	
Current assets		Current liabilities	
Cash and marketable sec.	2,600	Debt due for repayment	200
Accounts receivable	500	Accounts payable	9,900
Inventories	9,000	Total current liabilities	10,100
Total current assets	12,100	Long-term liabilities	11,300
Fixed assets		Total liabilities	21,400
Tangible fixed assets			
Property, plant, equipment	48,200	Shareholders' equity	
Less accum. depreciation	19,700	Paid-in capital	6,300
Net tangible fixed assets	28,500	Retained earnings	13,600
Intangible asset	700	Total shareholders' equity	19,900
Total fixed assets	29,200		
Total assets	**41,300**	**Total liabilities and equity**	**41,300**

Paid first ↓ Paid last

Financial Liquidity

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

Financial Liquidity

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}}$$

Example, Liquidity

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{12{,}100}{10{,}100} = 1.20$$

Example, Liquidity

$$\frac{\text{Current assets}}{\text{Current liabilities}} = \frac{12{,}100}{10{,}100} = 1.20$$

$$\frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}} = \frac{2{,}600 + 500}{10{,}100} = 0.31$$

Practice Problem

- Consider this simplified balance sheet for a business.

Balance Sheet ($)			
Current assets	$100	Current liabilities	$60
Long-term assets	500	Long-term debt	280
		Other liabilities	70
		Equity	190
Total	**$600**	**Total**	**$600**

a. What is the company's debt-to-equity ratio?

b. What is the ratio of total long-term debt to total long-term capital?

c. What is its current ratio?

Practice Problem

- Consider this simplified balance sheet for a business.

Balance Sheet ($)			
Current assets	$100	Current liabilities	$60
Long-term assets	500	Long-term debt	280
		Other liabilities	70
		Equity	190
Total	**$600**	**Total**	**$600**

a. What is the company's debt-to-equity ratio? $(\$60 + 280 + 70)/\$190 = 2.16$

b. What is the ratio of total long-term debt to total long-term capital?

c. What is its current ratio?

Practice Problem

- Consider this simplified balance sheet for a business.

Balance Sheet ($)			
Current assets	$100	Current liabilities	$60
Long-term assets	500	Long-term debt	280
		Other liabilities	70
		Equity	190
Total	**$600**	**Total**	**$600**

a. What is the company's debt-to-equity ratio? $(\$60 + 280 + 70)/\$190 = 2.16$

b. What is the ratio of total long-term debt to total long-term capital? $\$280/(\$280 + 190) = 0.60$

c. What is its current ratio?

Practice Problem

- Consider this simplified balance sheet for a business.

Balance Sheet ($)			
Current assets	$100	Current liabilities	$60
Long-term assets	500	Long-term debt	280
		Other liabilities	70
		Equity	190
Total	**$600**	**Total**	**$600**

a. What is the company's debt-to-equity ratio? $(\$60 + 280 + 70)/\$190 = 2.16$

b. What is the ratio of total long-term debt to total long-term capital? $\$280/(\$280 + 190) = 0.60$

c. What is its current ratio? $\$100/\$60 = 1.67$

Practice Problem

- A firm uses $1 million in cash to purchase inventories.

a. Will its current ratio rise or fall?

b. Will its quick ratio rise or fall?

Practice Problem

- A firm uses $1 million in cash to purchase inventories.

a. Will its current ratio rise or fall?
   The current ratio is unaffected. Inventories replace cash, but total current assets are unchanged.
b. Will its quick ratio rise or fall?

Practice Problem

- A firm uses $1 million in cash to purchase inventories.

a. Will its current ratio rise or fall?
   The current ratio is unaffected. Inventories replace cash, but total current assets are unchanged.
b. Will its quick ratio rise or fall?
   The quick ratio falls, since inventories are not included in the most liquid assets.

Practice Problem

- If a company has a healthy current ratio but a significantly lower quick ratio, then you can assume that,

a. the cost of goods sold represents more than half of revenues.
b. current liabilities exceed current assets.
c. the firm sells only on a cash basis.
d. inventory represents a large portion of the firm's current assets.

Practice Problem

- If a company has a healthy current ratio but a significantly lower quick ratio, then you can assume that,

a. the cost of goods sold represents more than half of revenues.
b. current liabilities exceed current assets.
c. the firm sells only on a cash basis.
d. inventory represents a large portion of the firm's current assets.

- The current ratio's numerator (current assets) includes inventories, the quick ratio's numerator does not

Practice Problem

- Which one of the following would be most detrimental to a firm's current ratio if that ratio is currently 2?

a. Collecting payment on an accounts receivable.
b. Selling marketable securities at cost.
c. Paying off accounts payable with cash.
d. Purchasing inventory on credit.

Practice Problem

- Which one of the following would be most detrimental to a firm's current ratio if that ratio is currently 2?

a. Collecting payment on an accounts receivable.
b. Selling marketable securities at cost.
c. Paying off accounts payable with cash.
d. Purchasing inventory on credit.

- The next slides work through each option with an example

Practice Problem

- We can think of an example,

Cash + marketable securities	0.5	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	0.5
Inventories	1		
Current assets	**2**	**Current liabilities**	**1**

Practice Problem

- We can think of an example,

Cash + marketable securities	0.5	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	0.5
Inventories	1		
Current assets	**2**	**Current liabilities**	**1**

- Current ratio = Current assets / Current liabilities = 2

Practice Problem

- Option a, cash rises and accounts receivable falls by the same amount, so current assets stay the same, and the current ratio stays the same

Practice Problem

- Option a, cash rises and accounts receivable falls by the same amount, so current assets stay the same, and the current ratio stays the same
- Option b, some marketable securities turn into cash, but cash + marketable securities stays the same, so the current ratio stays the same

Practice Problem

- Option c, suppose the firm pays down 0.5 of accounts payable, then

Cash + marketable securities	0	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	0
Inventories	1		
Current assets	**1.5**	**Current liabilities**	**0.5**

Practice Problem

- Option c, suppose the firm pays down 0.5 of accounts payable, then

Cash + marketable securities	0	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	0
Inventories	1		
Current assets	**1.5**	**Current liabilities**	**0.5**

- Current ratio = Current assets / Current liabilities = 3

Practice Problem

- Option d, suppose the firm acquires 1 of inventories on credit, then

Cash + marketable securities	0.5	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	1.5
Inventories	2		
Current assets	**3**	**Current liabilities**	**2**

Practice Problem

- Option d, suppose the firm acquires 1 of inventories on credit, then

Cash + marketable securities	0.5	Debt due for repayment	0.5
Accounts receivable	0.5	Accounts payable	1.5
Inventories	2		
Current assets	**3**	**Current liabilities**	**2**

- Current ratio = Current assets / Current liabilities = 1.5

Recap, Financing

- Financial leverage, debt raises shareholder returns in good times and lowers them in bad times
- Leverage ratios, how much debt the firm carries (debt ratios, times interest earned, cash coverage)
- Liquidity ratios, whether liquid assets can cover debt due soon (current and quick ratios)

Key Takeaways

- Shareholder value, the market-to-book ratio

Key Takeaways

- Shareholder value, the market-to-book ratio
- Investment, the returns on assets and equity, turnover ratios, and the profit margin

Key Takeaways

- Shareholder value, the market-to-book ratio
- Investment, the returns on assets and equity, turnover ratios, and the profit margin
- Financing, financial leverage, leverage ratios, and liquidity ratios

Key Takeaways

- Shareholder value, the market-to-book ratio
- Investment, the returns on assets and equity, turnover ratios, and the profit margin
- Financing, financial leverage, leverage ratios, and liquidity ratios
- Corporate performance is multi-dimensional, no single ratio captures it

Thank you!

Please let me know if you have any questions.